



FORCE-03

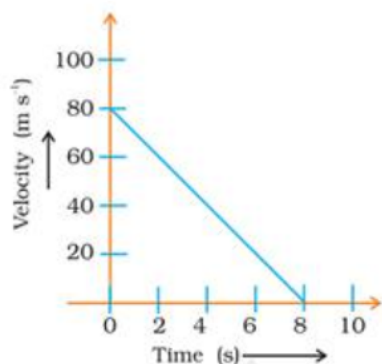
Class 09 - Science

1. An object experiences a net zero external unbalanced force. Is it possible for the object to be travelling with a non-zero velocity? If yes, state the conditions that must be placed on the magnitude and direction of the velocity. If no, provide a reason. [2]
2. Two blocks made of different metals identicals in shape and sizes are acted upon by equal forces which cause them to slide on a horizontal surface. The acceleration of the second block is found to be 5 times that of the first. What is the ratio of the mass of the second to the first? [2]
3. Deduce Newton's first law from the second law. [2]
4. A car of mass 1000 kg moving with a velocity of 45 km h^{-1} collides with a tree and comes to rest in 5s. What will be the force exerted by the car on the tree? [2]
5. From the rifle of mass 4 kg, a bullet of mass 50 g is fired with an initial velocity of 35 ms^{-1} . Calculate the initial recoil velocity of the rifle. [2]
6. Water sprinkler used for grass lawns begins to rotate as soon as the water is supplied. Explain the principle on which it works. [2]
7. A body of mass 500 g is at rest on a frictionless surface. Calculate the distance travelled by it in 10 s when acted upon by a force of 10^{-2} N . [2]
8. A horse continues to apply a force in order to move a cart with a constant speed. Explain why? [2]
9. It is required to increase the velocity of a scooter of mass 80 kg from 5 to 25 ms^{-1} in 2 second. Calculate the force required. [2]
10. Explain, why is it difficult for a fireman to hold a hose, which ejects large amounts of water at a high speed. [2]
11. Derive the unit of force using the second law of motion. A force of 5 N produces an acceleration of 8 ms^{-2} on a mass ' m_1 ' and an acceleration of 24 ms^{-2} on a mass ' m_2 '. What acceleration would the same force provide if both the masses are tied together? [2]
12. Give a few examples of Newton's third law of motion. [2]
13. Explain, why is it difficult for a fireman to hold a hose, which ejects large amounts of water at a high velocity. [2]
14. A hockey ball of mass 200 g travelling at 10 ms^{-1} is struck by a hockey stick so as to return it along its original path with a velocity at 5 ms^{-1} . Calculate the magnitude of change of momentum occurred in the motion of the hockey ball by the force applied by the hockey stick. [2]
15. An object undergoes an acceleration of 8 ms^{-2} starting from rest. Find the distance travelled in 1 second. [2]
16. Why do the driver and the person seated in front seat need a seat belt? [2]
17. A constant force acts on an object of mass 5 kg for a duration of 2 s. It increases the object's velocity from 3 ms^{-1} to 7 ms^{-1} . Find the magnitude of the applied force. Now, if the force was applied for a duration of 5 s, what [2]

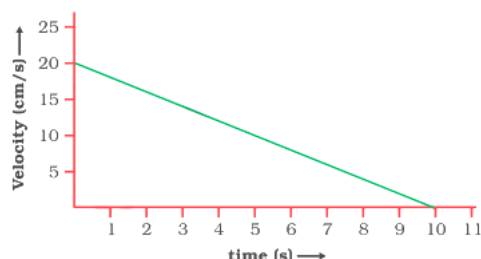
would be the final velocity of the object?

18. A bullet of mass 10 g is fired with a rifle. The bullet takes 0.003 s to move through its barrel and leaves with a velocity of 300 ms^{-1} . What is the force exerted on the bullet by the rifle? [2]
19. Why it is advised to tie any luggage kept on the roof of a bus with a rope? [2]
20. There are three solids balls made up of aluminium, steel and wood, of the same shape and same volume. Which of them would have highest inertia? [2]
21. Why do you fall in the forward direction when a moving bus brakes to a stop and fall backward when it accelerates from rest? [2]
22. An automobile vehicle has a mass of 1500 kg. What must be the force between the vehicle and road if the vehicle is to be stopped with the negative acceleration of 1.7 ms^{-2} ? [2]
23. A javelin throw is marked foul if an athlete crosses over the line marked for throw. Explain why the athletes often fall to stop themselves before the line. [2]
24. Describe our walking in terms of Newton's third law of motion. [2]
25. Two balls of the same size of different materials, rubber and iron are kept on the smooth floor of a moving train. The brakes are applied suddenly to stop the train. Will the balls start rolling? If so, in which direction? Will they move with the same speed? Give reasons for your answer. [2]
26. Describe balanced forces with the help of two examples. [2]
27. Why does an athlete run a certain distance before taking a leap? [2]
28. Why does a cricket player move his hands backward while catching the ball? [2]
29. Why can a small mass such as a bullet kill a person when fired from a gun? [2]
30. On what factors do the following physical quantities depend? [2]
 - i. Inertia
 - ii. Momentum
 - iii. Force
31. A bullet fired against a glass window pane makes a hole in it, and the glass pane is not cracked. But on the other hand, when a stone strikes the same glass pane, it gets smashed. Why is it so? [2]
32. A stone of 1 kg is thrown with a velocity of 20 ms^{-1} across the frozen surface of the lake and comes to rest after travelling a distance of 50 m. What is the force of friction between the stone and the ice? [2]
33. Explain why some of the leaves may get detached from a tree if we vigorously shake its branch. [2]
34. Explain why some of the leaves may get detached from a tree if we vigorously shake the branch of the tree. [2]
35. A motorcar is moving with a velocity of 108 km/h and it takes 4 s to stop after the brakes are applied. Calculate the force exerted by the brakes on the motorcar if its mass along with the passengers is 1000 kg. [2]
36. In the following example, try to identify the number of times the velocity of the ball changes:
"A football player kicks a football to another player of his team who kicks the football towards the goal. The goalkeeper of the opposite team collects the football and kicks it towards a player of his own team".
Also identify the agent supplying the force in each case. [2]
37. What will be acceleration of a body of mass 5 kg, if a force of 200 N is applied on it? [2]
38. A man throws a ball of mass 0.4 kg vertically upwards with a velocity of 10 m/s. What will be its initial momentum? What would be its momentum at the highest point of its reach? [2]
39. Explain why the sole of a shoe is worn out. [2]
40. A truck starts from rest and rolls down a hill with a constant acceleration. It travels a distance of 400 m in 20 s. Find its acceleration. Find the force acting on it if its mass is 7 metric tonnes. [2]

41. A hammer of mass 500 g, moving at 50 ms^{-1} , strikes a nail. The nail stops the hammer in a very short time of 0.01 s. What is the force of the nail on the hammer? [2]
42. A bullet of mass 10 g travelling horizontally with a velocity of 150 ms^{-1} strikes a stationary wooden block and comes to rest in 0.03 s. Calculate a distance of penetration of the bullet into the block. Also calculate the magnitude of the force exerted by the wooden block on the bullet. [2]
43. When a carpet is beaten with a stick, dust comes out of it. Explain. [2]
44. From a rifle of mass 4 kg, a bullet of mass 50 g is fired with an initial velocity of 35 ms^{-1} . Calculate the initial recoil velocity of the rifle. [2]
45. Two persons manage to push a motor car of mass 1,200 kg at a uniform velocity along a level road. The same motor car can be pushed by three persons to produce an acceleration of 0.2 ms^{-2} . With what force does each person push the motor car? Assume that all persons push the motor car with the same muscular effort. [2]
46. A truck of mass M is moved under a force F. If the truck is then loaded with an object equal to the mass of the truck and the driving force is halved, then how does the acceleration change? [2]
47. A truck starts from rest and rolls down a hill with constant acceleration. It travels a distance of 400 metres in 20 s. Find its acceleration. Find the force acting on it, if its mass is 7 metric tons. [2]
48. Velocity versus time graph of a ball of mass 50 g rolling on a concrete floor is shown in Fig. Calculate the acceleration and frictional force of the floor on the ball. [2]



49. When a force of 40 N is applied on a body it moves with an acceleration of 5 ms^{-2} . Calculate the mass of the body. [2]
50. Which would require a greater force, accelerating a 2 kg mass at 5 ms^{-2} or a 4 kg mass at 2 ms^{-2} ? [2]
51. The velocity-time graph of a ball of mass 20 g moving along a straight line on a long table is given in Fig. [2]



How much force does the table exert on the ball to bring it to rest?

52. Why does a boat tend to leave the shore, when passengers are alighting from it? [2]
53. A force of 5 N gives a mass m_1 , an acceleration of 10 ms^{-2} and a mass m_2 , an acceleration of 20 ms^{-2} . What acceleration would it give if both the masses were tied together? [2]
54. A bullet of mass 5 g travelling at a speed of 120 ms^{-1} penetrates deeply into the fixed target and is brought to rest in 0.01 s. Calculate: (a) the distance of penetration in the target, (b) the average force exerted on the bullet. [2]
55. A hammer of mass 500g, moving at 50 ms^{-1} , strikes a nail. The nail stops the hammer in a very short time of [2]

- 0.01s. What is the force of the nail on the hammer?
56. A batsman hits a cricket ball which then rolls on a level ground. After covering a short distance, why does the ball come to rest? [2]
 57. In the oil tankers some space is left at the top while filling them. Explain. [2]
 58. Two balls of the same size of different materials, rubber and iron are kept on the smooth floor of a moving train. The brakes are applied suddenly to stop the train. Will the balls start rolling? If so, in which direction? Will they move with the same speed? Give reasons for your answer. [2]
 59. Explain why some of the leaves may fall from a tree, if we vigorously shake its branch. [2]
 60. Describe in brief an activity to illustrate the property of inertia of rest. [2]
 61. An automobile vehicle has a mass of 1500 kg. What must be the force between the vehicle and road if the vehicle is to be stopped with a negative acceleration of 1.7 ms^{-2} ? [2]
 62. A 8000 kg engine pulls a train of 5 wagons, each of 2000 kg, along a horizontal track. If the engine exerts a force of 40000 N and the track offers a friction force of 5000 N, then calculate: [2]
 - a. the net accelerating force and
 - b. the acceleration of the train.
 63. According to the third law of motion, when we push on an object, the object pushes back on us with an equal and opposite force. If the object is a massive truck parked along the roadside, it will probably not move. A student justifies this by answering that the two opposite and equal forces cancel each other. Comment on this logic and explain why the truck does not move. [2]
 64. Which of the following has more inertia: [2]
 - a. a rubber ball and a stone of the same size?
 - b. a bicycle and a train?
 - c. a five-rupee coin and a one-rupee coin?
 65. A stone of 1 kg is thrown with a velocity of 20 ms^{-1} across the frozen surface of a lake and comes to rest after travelling a distance of 50 m. What is the force of friction between the stone and the ice? [2]
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 - a. a rubber ball and a stone of the same size?
 - b. a bicycle and a train?
 - c. a five rupees coin and a one-rupee coin?
 67. A motorcar of mass 1200 kg is moving along a straight line with a uniform velocity of 90 km/h. Its velocity is slowed down to 18 km h^{-1} in 4 s by an unbalanced external force. Calculate the acceleration and change in momentum. Also, calculate the magnitude of the force required. [2]
 68. Why do you fall in the forward direction when a moving bus brakes to a stop and fall backwards when it accelerates from rest? [2]
 69. Using second law of motion, derive the relation between force and acceleration. A bullet of mass 10 g strikes a sand bag with a velocity of 10^3 ms^{-1} and gets embedded after travelling 5 cm. Calculate [2]
 - i. the resistive force exerted by the sand bag on the bullet.
 - ii. the time taken by the bullet to come to rest.
 70. How much momentum will a dumbbell of mass 10 kg transfer to the floor, if it falls from a height of 0.8 m? [2]
Take acceleration due to gravity as 10 ms^{-2} .
 71. State Newton's third law of motion and how does it explain the walking of man on the ground? [2]

72. An iron sphere of mass 1 kg is dropped from a height of 10 m. If the acceleration of sphere is 9.8 ms^{-2} , calculate the momentum transferred to the ground by the ball. [2]
73. Why is it difficult to balance our body, when we accidentally step on a peel of banana? [2]